

Final exam Quantum Physics (GEMF)

Coen de Graaf — Departament de Química Física i Inorgànica, URV

- The exam has 6 questions, each question is worth $\frac{5}{3}$ points.
- Pages are printed on both sides.
- Answer the questions on separate sheets, clearly mark each sheet with your name.
- When asked for a numerical answer, always demonstrate how you got the number. Just giving the number will not be enough to get the maximum points.

Good luck

1. Stern Gerlach experiment



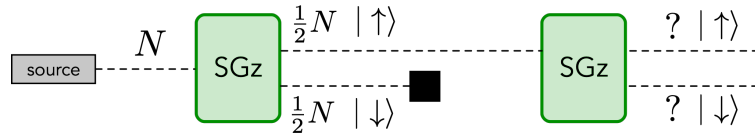
The left side of the picture above gives a schematic representation of the Stern-Gerlach experiment. The **source** sends electrons in the quantum state

$$|\text{source}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$$

through a magnetic field along the z -axis (**SGz**) to give rise to two possible outcomes, one with the electron in the $|\uparrow\rangle$ quantum state and one with the electron in the $|\downarrow\rangle$ quantum state. The right side of the picture is similar but now the magnetic field is aligned along the x -axis (**SGx**). The electrons that leave the magnetic field are either in quantum state $|+\rangle = (|\uparrow\rangle + |\downarrow\rangle)/\sqrt{2}$ or $|-\rangle = (|\uparrow\rangle - |\downarrow\rangle)/\sqrt{2}$.

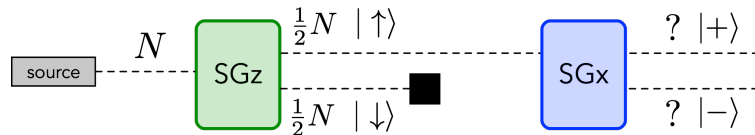
- (a) Write down the quantum states $|\text{source}\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|+\rangle$ and $|-\rangle$ as column vectors.

In a new experimental setup, the electrons in quantum state $|\downarrow\rangle$ are blocked by the black square in the figure below and only the electrons in the $|\uparrow\rangle$ quantum state reach the second magnetic field, which is also aligned along the z -axis.



- (b) When the source emits N electrons, how many electrons will leave the second magnetic field in quantum state $|\uparrow\rangle$ and how many in quantum state $|\downarrow\rangle$?

In the next experiment, schematically represented in the figure below, the electrons in quantum state $|\uparrow\rangle$ are sent through a magnetic field aligned along the x -axis.



- (c) How many electrons will leave the second magnetic field in quantum state $|+\rangle$ and how many in quantum state $|-\rangle$?

2. Operators and measurements

Given the matrix representations of the operators $\hat{A}$ and $\hat{B}$

$$\hat{A} = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix} \quad \hat{B} = \begin{pmatrix} 1 & \sqrt{8} & 0 \\ \sqrt{8} & -1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

- (a) Show that both operators are Hermitian.
- (b) Derive the eigenvalues of $\hat{A}$ and $\hat{B}$.
- (c) Given that the system at $t = 0$ is described by

$$|\psi\rangle = N \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

calculate the expectation values of $\hat{A}$ and $\hat{B}$ (don't forget to calculate N first)

- (d) What is the probability of measuring $A = \sqrt{5}$ at $t = 0$?
- (e) What is the probability of measuring $B = 2$ at $t = 0$?
- (f) What is the probability of first measuring $A = \sqrt{5}$ and immediately thereafter $B = 2$?

The eigenfunctions of $\hat{A}$ and $\hat{B}$ are

$$\begin{aligned} \phi_{A=\sqrt{5}} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{-i}{\sqrt{5}} \\ 1 \\ \frac{2}{\sqrt{5}} \end{pmatrix} & \phi_{A=-\sqrt{5}} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{i}{\sqrt{5}} \\ 1 \\ \frac{2}{\sqrt{5}} \end{pmatrix} & \phi_{A=0} &= \frac{1}{\sqrt{5}} \begin{pmatrix} 2i \\ 0 \\ -1 \end{pmatrix} \\ \phi_{B=3} &= \frac{1}{\sqrt{3}} \begin{pmatrix} \sqrt{2} \\ 1 \\ 0 \end{pmatrix} & \phi_{B=-3} &= \frac{1}{\sqrt{3}} \begin{pmatrix} -1 \\ \sqrt{2} \\ 0 \end{pmatrix} & \phi_{B=2} &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \end{aligned}$$

3. Angular momentum algebra

Consider a particle with angular momentum $j = 1$ whose Hamiltonian is given by

$$\hat{H} = \frac{\lambda}{\hbar^2}(\hat{J}_x^2 + \hat{J}_y^2 - \hat{J}_z^2) + 2\frac{\mu}{\hbar}\hat{J}_z$$

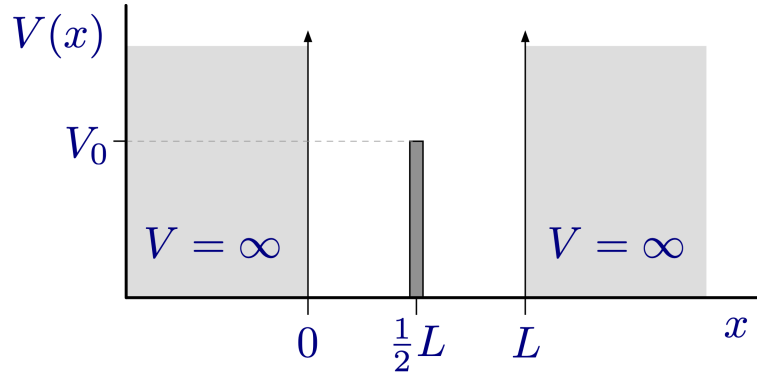
where λ and μ are constants, and the quantum state is written as $|j, m\rangle$.

- (a) How many different quantum states (being eigenfunctions of $\hat{J}^2$ and $\hat{J}_z$) can the particle adopt? Use the $|j, m\rangle$ notation to specify quantum states.
- (b) Find the energy levels.
(Hint: rewrite the Hamiltonian using $\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$.)
- (c) Are the energy levels degenerate?

$$\hat{J}^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle \quad \hat{J}_z |j, m\rangle = \hbar m |j, m\rangle$$

4. Perturbation theory

- (a) Derive the expression for the wave functions that describe the quantum state of a particle that is confined in a one-dimensional box of length L . The potential is zero between 0 and L , and infinite for $x > L$ and $x < 0$.
- (b) Calculate the first-order correction to the ground state energy of the particle after adding a constant potential V_0 in the interval $\frac{1}{2}L - a < x < \frac{1}{2}L + a$. Repeat the exercise for the first excited state. Note that the width of the extra potential is small enough to assume that the wave function is constant in the region of the perturbation.



The unperturbed system is described by

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi^{(0)}}{dx^2} = E^{(0)} \psi^{(0)}$$

The general solution of the differential equation

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0 \quad \text{can be written as} \quad \psi(x) = A \sin(kx) + B \cos(kx)$$

The first order energy correction is given by the equation

$$E_n^{(0)} = \langle \psi_n^{(0)} | \hat{V} | \psi_n^{(0)} \rangle$$

5. Harmonic oscillator

- (a) Use the variational theory to find the optimal value of α to describe the ground state of the harmonic oscillator with trial function $\psi_\alpha = Ae^{-\alpha x^2}$
- (b) Compare to the exact ground state energy and based on this comparison, comment the quality of the trial wave function

The Hamiltonian and the exact ground state energy are given by

$$\hat{H} = \frac{-\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2 \quad E_0(\text{exact}) = \frac{1}{2}\hbar\omega$$

Use the following standard integrals:

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \quad \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \frac{1}{2a} \sqrt{\frac{\pi}{a}}$$

Normalization implies

$$\int_{-\infty}^{\infty} \psi_\alpha^* \psi_\alpha dx = \int_{-\infty}^{\infty} A^2 e^{-2\alpha x^2} dx = 1$$

6. Comparing Bohr's model of the hydrogen atom to the exact solution

(a) Complete the Table for Bohr's model of the hydrogen atom

n	E_n (eV)	r_n (Å)
1	...	...
2	...	...
3	...	...

(b) Complete the Table for the exact solutions of the hydrogen atom. P_{max} refers to the highest maximum in the probability distribution. For R_{10} , R_{21} and R_{32} , the exact value must be calculated, for the other three quantum states an estimate based on graphical inspection suffices (see graph on the next page).

R_{nl}	E_n (eV)	P_{max} (Å)
R_{10}	...	...
R_{20}	...	...
R_{21}	...	...
R_{30}	...	...
R_{31}	...	...
R_{32}	...	...

(c) Based on these data, shortly discuss the coincidences and shortcomings of Bohr's model of the hydrogen atom (energies, degeneracy, completeness of the model, Bohr radii versus maximum in the probability functions, ...)

$$\begin{aligned}
 R_{10}(r) &= 2e^{-r} & R_{20}(r) &= \frac{1}{\sqrt{2}}(1 - \frac{1}{2}r)e^{-r/2} \\
 R_{30}(r) &= \frac{2}{3\sqrt{3}}(1 - \frac{2}{3}r + \frac{2}{27}r^2)e^{-r/3} & R_{21}(r) &= \frac{1}{2\sqrt{6}}re^{-r/2} \\
 R_{31}(r) &= \frac{8}{9\sqrt{6}}(\frac{1}{3}r - \frac{1}{18}r^2)e^{-r/3} & R_{32}(r) &= \frac{4}{81\sqrt{30}}r^2e^{-r/3}
 \end{aligned}$$

Bohr model:

$$E_n = -\frac{m_e}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} \quad r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2$$

$$1 \text{ au} = 27.211 \text{ eV}$$

$$1 \text{ au} = 0.529177 \text{ Å}$$

$$\hbar = 1 \text{ au} ; m_e = 1 \text{ au} ; 4\pi\epsilon_0 = 1 \text{ au} ; e = 1 \text{ au}$$

